

SignalScope

Risk intelligence for wholesale portfolios

AI/NLP risk engine and strategic portfolio stress testing

Problem and approach

The gap

Financial news and social posts arrive as unstructured text. Analysts need to identify the event, judge its severity, and connect it to portfolio exposures.

SignalScope

- 01** Ingest public news and social text
- 02** Publish sentiment, event class and impact
- 03** Stress a synthetic wholesale portfolio

Output: an inspectable risk signal and an illustrative portfolio loss estimate

System design

GDELT / Google News RSS headlines, Bluesky posts and replay records share one signal contract.



One risk signal links a source item to a repeatable stress view

NLP scoring and stress mechanics

AI/NLP risk engine

Sentiment	Finance lexicon with negation, bounded from -1 to +1
Event	Multinomial Naive Bayes on curated phrases, with domain cues
Impact	Ordinal 1–10 rubric from event class and severity language

Portfolio stress model

Trigger	A high-impact signal (score >7) selects an event shock vector
Portfolio	12 synthetic positions: 5 loans, 4 bonds, 3 derivatives
Valuation	Duration approximations and signed delta/PV01 sensitivities

Simulated credit stress: rates +20 bp, broad spreads +55 bp, Industrials spread +140 bp, Aster haircut 25%.

Credit event stress result

Signal: Credit Event, impact 10/10, sentiment -0.990

Bundled simulated headline: Aster Manufacturing misses bond coupon and warns of default.

Before stress

\$797.00m

After stress

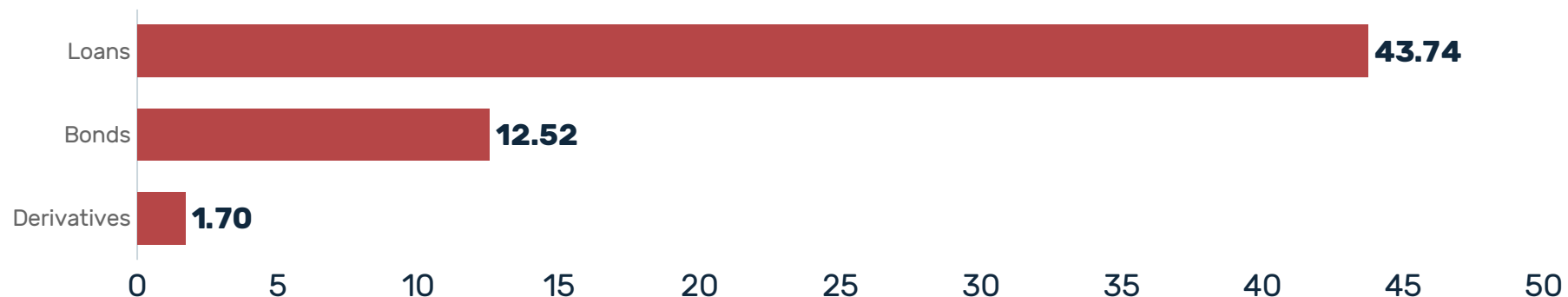
\$739.04m

Scenario loss

\$57.96m

7.27% of starting value

Loss contribution by asset class (USD m)



Simulated USD m. Synthetic positions and illustrative scenario shocks.

Domain impact and next steps

A first-pass risk view

One event record gives analysts the source text, a structured severity signal, and an illustrative change in portfolio value.

Risk teams can prioritize review and see which asset class drives the scenario loss.

Challenges to validate

- 01** Public feeds vary in coverage and credibility.
- 02** Text scores need evaluation on labeled finance data.
- 03** Scenario shocks need calibration to market and credit data.

Next: evaluate signal quality, then calibrate shocks before decision use